Demo Plan & Future Roadmap

FROM LIVE DEMO TO LASTING IMPACT

Build Today for **Stronger Tomorrows**

1 LIVE DEMO STRATEGY



Presenter wears the glove on stage and performs correct grip-and-release reps

Dashboard shows live flex + EMG signal trace and rep counter in real time

A deliberately incomplete rep is performed to trigger haptic/visual deviation feedback

Session summary + multi-session trend chart shown to close the demo

2 EXPECTED DELIVERABLES

- Working glove prototype (flex + EMG sensing, ESP32 acquisition)
- Real-time rep counting with correct/deviation classification
- Live companion dashboard (signal view, rep counter, session summary)
- Self-collected labeled dataset + trained classifier with evaluation metrics

3 FUTURE SCOPE & ROADMAP

SHORT TERM

- Add 2+ finger sensing, BLE streaming polish, per-user calibration
- Improve dashboard: live trace + session history

MID TERM

- Multiple exercise types beyond grip-and-release
- IMU integration to detect compensatory movement
- On-device ML inference — no companion app needed

LONG TERM

- Dedicated mobile app + cloud sync
- Therapist-facing analytics dashboard
- Clinical validation study with real patient population

0 – 6 Months → **6 – 18 Months** → **18+ Months**